Lecture 10 (10/7)

Confidence intervals and least squares

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Today:

- Unit 3.3/3.6: Confidence intervals, using the estimated sample distribution, and hypothesis testing.
- Unit 3.4: Least squares estimators of $\beta_0, \beta_1, \sigma_\epsilon^2$ and R^2, and the sample distribution of $\hat{\beta}_1$.

True vs. estimated sample distribution

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(red: numbers computed from our data)

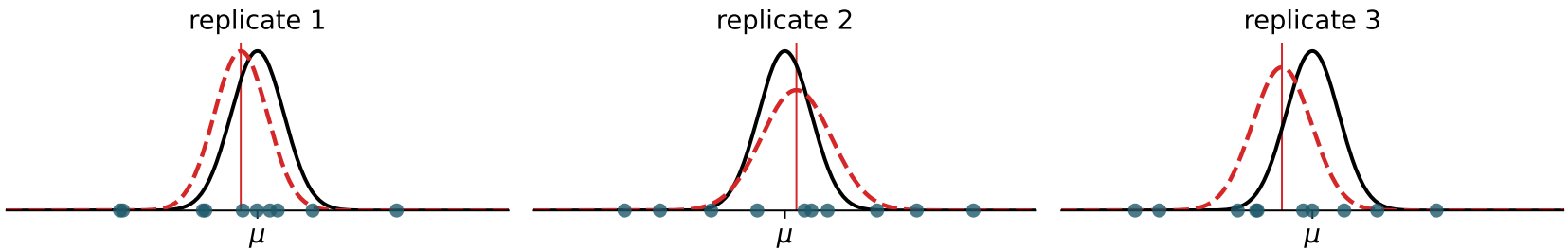
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Points: data ($N = 10$). Black: true. Red dashed: estimated, centered at $\hat{\mu}$.

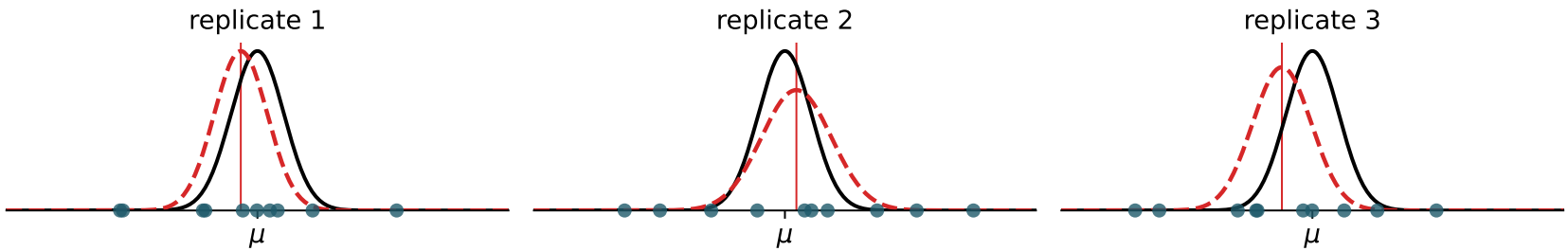
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We need to estimate the parameters in the sample distribution, which means the entire distribution is itself a random variable. Still, its width tells us how far μ might plausibly be from $\hat{\mu}$. This idea is formalized with confidence intervals.

Confidence intervals

The 95% confidence interval:

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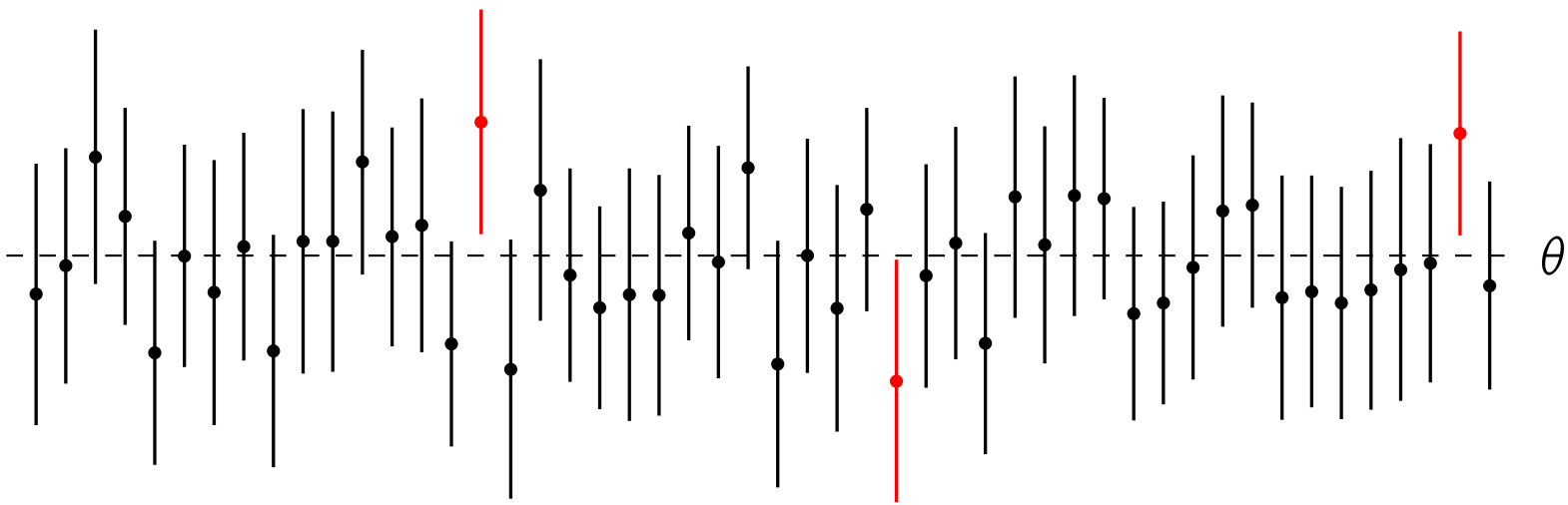
For an approximately Normal sample distribution, 95% of values fall within 1.96 standard errors of the center. (With $\hat{\sigma}$ plugged in, the exact multiplier comes from a *t* distribution: 2.26 for $N = 10$, 2.05 for $N = 30$, ≈ 1.96 for large N.)

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Each bar is the CI from one replicate, with a dot at $\hat{\theta}$. About 95% contain θ; the red ones miss.

Subtleties of interpretation (optional)

If you want to get pedantic about CIs, there is a distinction which is more about the language we use to talk about them than how they are used practically:

- **Correct:** if we generate many replicates of the data, the true θ falls in the resulting CI for 95% of them; that is,

$$P(\theta \in \mathrm{CI}) = .95$$

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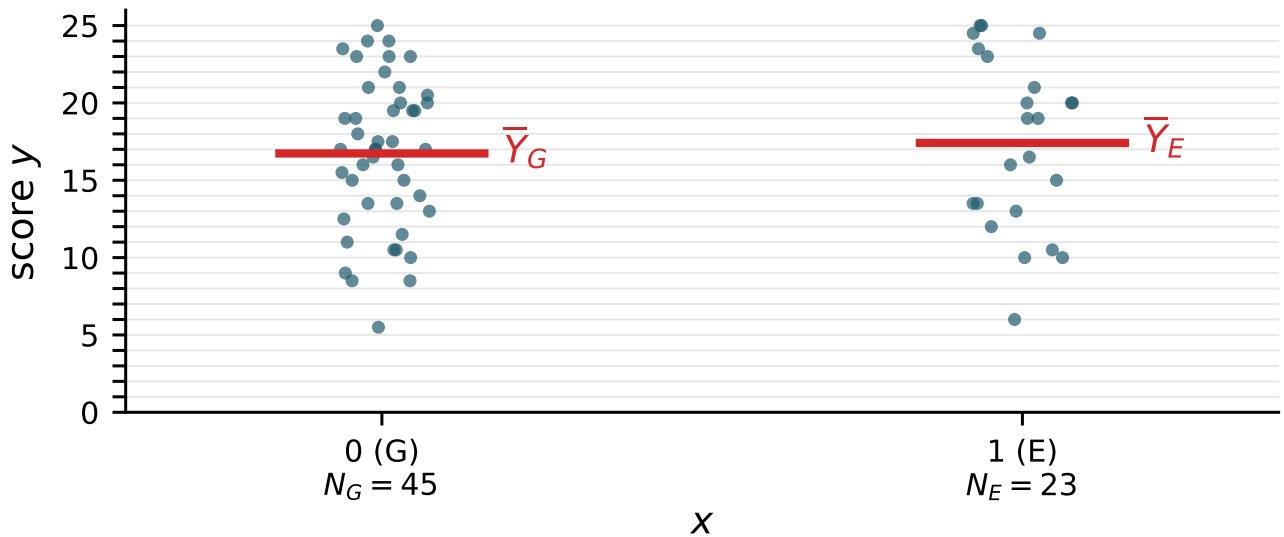
- **Not correct:** there's a 95% chance θ lies in *this particular* interval. Once we compute, say, $\mathrm{CI} = [0.2, 1.4]$, θ is either in it or not:

$$P(\theta \in \mathrm{CI}) = 0 \quad \text{OR} \quad P(\theta \in \mathrm{CI}) = 1$$

Question

Midterm 1 was graded by two graders, E and G. Let Y be the score (out of 25) and $X = 1$ if E graded the exam ($X = 0$ if G did). We model the scores as

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X,\ \sigma_\epsilon^2).$$



- Compute $\overline{Y|X=0}$, $\overline{Y|X=1}$, $\hat{\beta}_1$ and $\hat{\sigma}_\epsilon^2$.
- What is the sample distribution of $\hat{\beta}_1$?

Sample distribution of a difference of means

An "eyeball" calculation

$$\overline{Y|X=0} \approx 17, \quad \overline{Y|X=1} \approx 17.5, \quad \hat{\sigma}_\epsilon \approx 5$$

hence

$$\overline{Y|X=1} \approx \text{Normal}(17.5, 25/23), \quad \overline{Y|X=0} \approx \text{Normal}(17, 25/45),$$

and therefore using $25/23 + 25/45 \approx 1 + 1/2 = 3/2$,

$$\hat{\beta}_1 = \overline{Y|X=1} - \overline{Y|X=0} \approx \text{Normal}(1/2, 3/2)$$

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In Python (with `x`, `y` numpy arrays):

```
y0 = y[x == 0]
y1 = y[x == 1]
b1_hat = np.mean(y1) - np.mean(y0)
var_b1 = np.std(y1)**2/len(y1) + np.std(y0)**2/len(y0)
```

gives us

$$\hat{\beta}_1 = \overline{Y|X=1} - \overline{Y|X=0} \approx \text{Normal}(0.68,\ 1.84).$$

Hypothesis test

Question

Using "eyeball" numbers

- Find the 95%-CI for β_1.
- Hypothesis test: if the null hypothesis $H_0 : \beta_1 = 0$ (same grading) were true, the test statistic $\hat{T}^* = \hat{\beta}_1^*/\text{se}(\hat{\beta}_1) \approx \textsf{Normal}(0, 1)$. To decide whether we can reject this hypothesis, we compute the p-value $P(|\hat{T}^*| > |\hat{T}|)$.

Hypothesis test

- We already found $\text{se}(\hat{\beta}_1) \approx \sqrt{3/2} \approx 1.2$, so (using $1.96 \approx 2$) $\text{CI} \approx [1/2 - 2(1.2),\ 1/2 + 2(1.2)] \approx [-2,\ 3]$, which contains 0.

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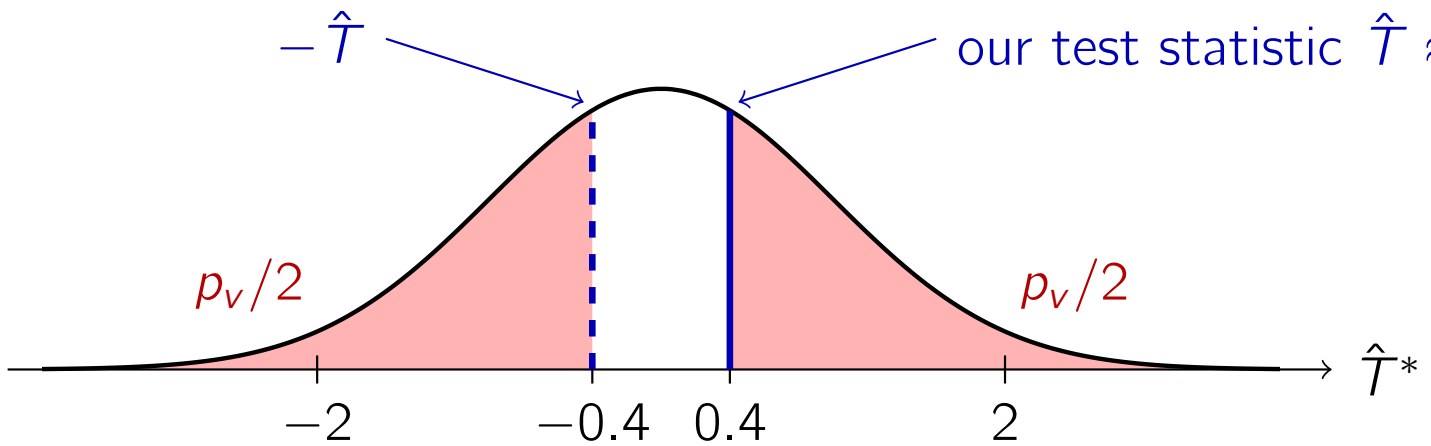
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- Near the center the Normal density is about 0.4, so $P(|\hat{T}^*| < 0.4) \approx 0.8 \times 0.4 \approx 0.3$ (a rectangle under the bell curve), hence $p_v = P(|\hat{T}^*| > 0.4) \approx 0.7$. A difference like ours is typical if the graders grade the same, so we *cannot* reject H_0.

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Estimators of regression parameters

Data $(X_1, Y_1), \ldots, (X_N, Y_N)$ from $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$. How do we estimate β_1?

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Replacing each expectation with a sample average, $E[\,\cdot\,] \to \frac{1}{N}\sum_i$, yields

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}, \qquad \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}.$$

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The residuals are $r_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$, and we define

$$\mathrm{SSR} = \sum_i r_i^2 \qquad \text{(sum of squared residuals)},$$

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The estimators of the noise variance and of ρ^2 are

$$\boxed{\hat{\sigma}_\epsilon^2 = \frac{\mathrm{SSR}}{N-2} \approx \frac{\mathrm{SSR}}{N} \qquad \hat{\rho}^2 = R^2 = 1 - \frac{\mathrm{SSR}}{\mathrm{SST}}}$$

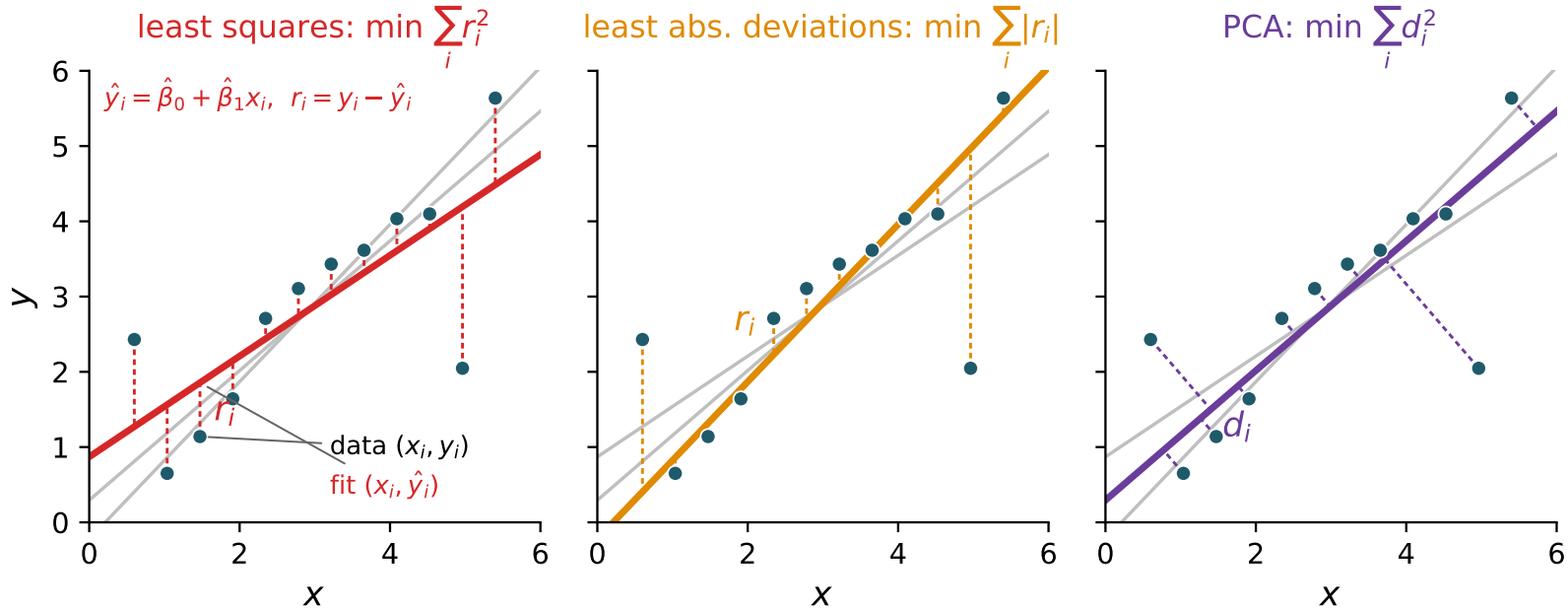
($N-2$ makes the estimator unbiased, but makes little difference once $N \gtrsim 20$)

Least squares

$\hat{\beta}_0, \hat{\beta}_1$ give the line with the smallest SSR,

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{b_0,\, b_1}{\operatorname{argmin}} \sum_i \left(Y_i - b_0 - b_1 X_i\right)^2,$$

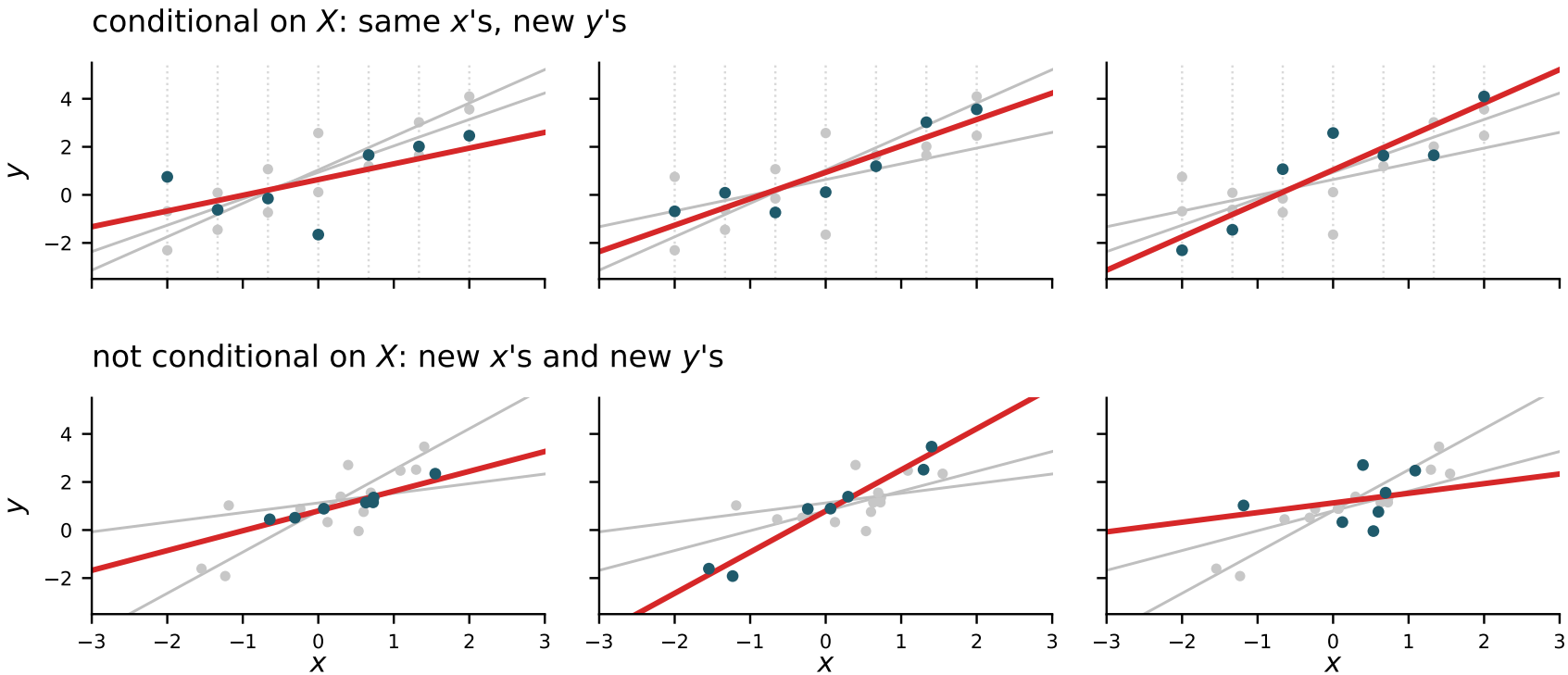
which is why they are called the least squares estimators. Other choices give other lines:



Same data, three lines (others in gray). LS and least abs. deviations use vertical residuals r_i; PCA uses perpendicular distances d_i. Squaring makes LS sensitive to the outlier.

Two kinds of replicates

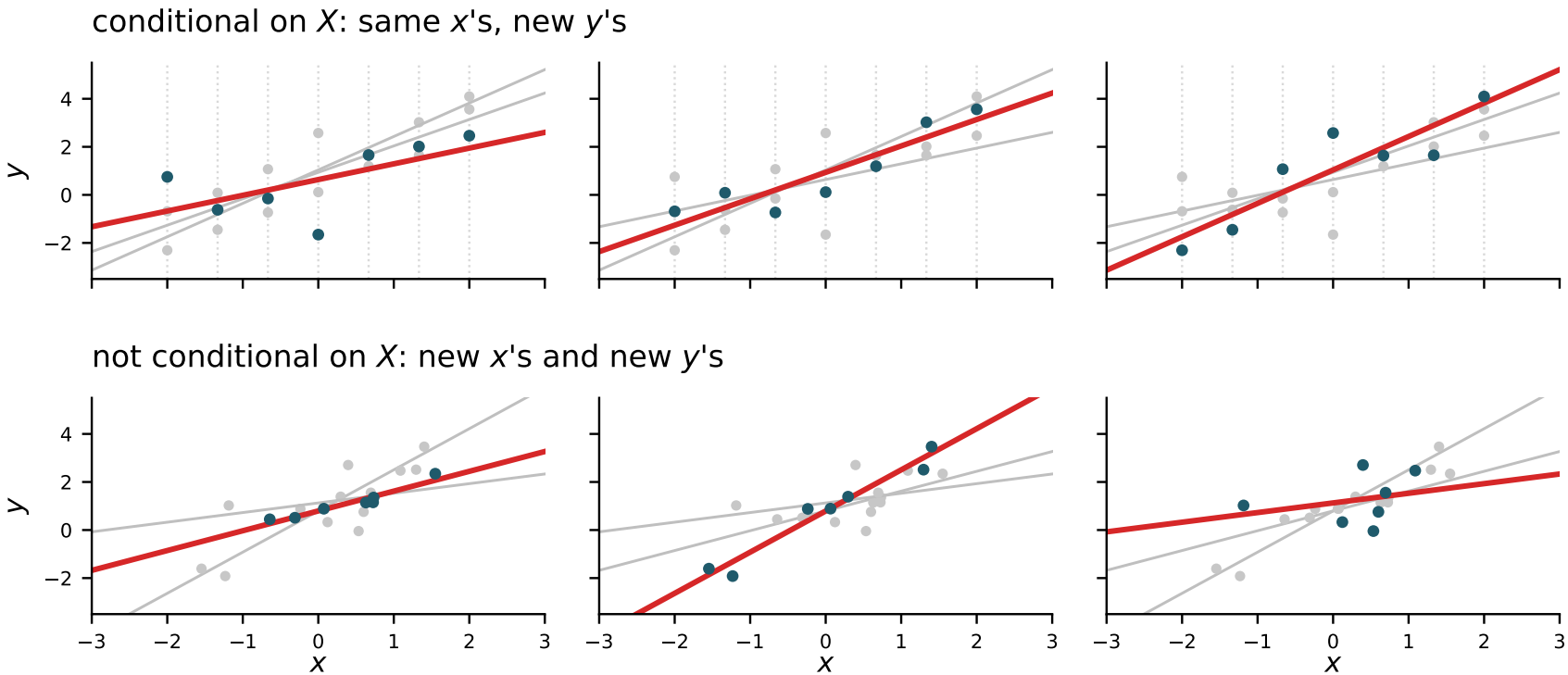
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We focus on the sample distribution conditioned on $X_1, \ldots, X_N$.

$\hat{\beta}_1$ is an unbiased estimator

$$E[\hat{\beta}_1|X_1,\ldots,X_N] = \frac{\sum_{i=1}^{N}(X_i-\overline{X})E[(Y_i-\overline{Y})|X_1,\ldots,X_N]}{\sum_{i=1}^{N}(X_i-\overline{X})^2}$$

and

$$E[(Y_i-\overline{Y})|X_1,\ldots,X_N] = \beta_0+\beta_1X_i-\beta_0-\beta_1\overline{X} = \beta_1(X_i-\overline{X})$$

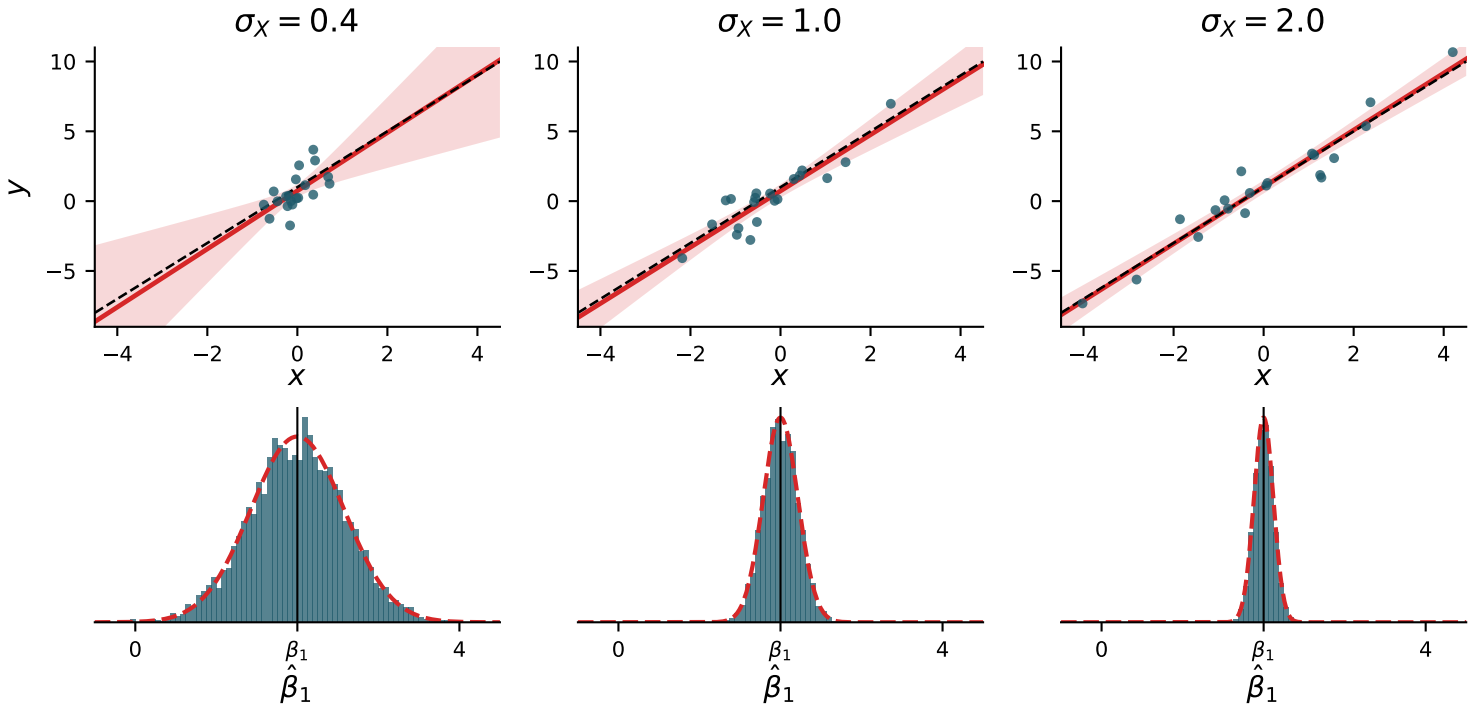
so

$$E[\hat{\beta}_1|X_1,\ldots,X_N] = \beta_1\frac{\sum_{i=1}^{N}(X_i-\overline{X})^2}{\sum_{i=1}^{N}(X_i-\overline{X})^2} = \beta_1$$

Sample distribution of $\hat{\beta}_1$

For large N,

$$\hat{\beta}_1 \approx \text{Normal}\left(\beta_1, \frac{\sigma_\epsilon^2}{N\sigma_X^2}\right)$$



$N = 20$, $\beta_1 = 2$, $\sigma_\epsilon = 1$. Top: one dataset, its fitted line (red), the 95% interval for the line (shaded) and the true line (dashed). Bottom: $\hat{\beta}_1$ over 5000 replicates with the same x's, with the Normal approximation (dashed).

Back to the graders

- Recall the graders example: $N = 68$ exams, $N_E = 23$ graded by E, and $\hat{\sigma}_\epsilon \approx 5$.
- Here $X \sim \text{Bernoulli}(p)$ with $p \approx N_E/N = 23/68 \approx 1/3$, and

$$\sigma_X^2 = p(1-p) \approx 2/9 \implies N\sigma_X^2 \approx 68 \times \frac{2}{9} = \frac{136}{9} \approx 15.1.$$

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- So $\hat{\beta}_1 \approx \text{Normal}\left(\beta_1,\ 1.65\right)$, since

$$\text{var}(\hat{\beta}_1) \approx \frac{\hat{\sigma}_\epsilon^2}{N\sigma_X^2} \approx \frac{25}{136/9} = \frac{225}{136} \approx 1.65, \qquad \text{se}(\hat{\beta}_1) \approx \sqrt{1.65} \approx 1.29.$$

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- Compare with what we found before:
  - back of the envelope: $\text{var}(\hat{\beta}_1) \approx 3/2$, $\text{se}(\hat{\beta}_1) \approx 1.2$;
  - exact (Python): $\text{var}(\hat{\beta}_1) = 1.84$, $\text{se}(\hat{\beta}_1) \approx 1.36$.

  All three roughly agree: the boxed formula gives the same answer as the difference of means.